Chapter 2

Static and Dynamic Characteristics of Signals

2.1 INTRODUCTION

A measurement system takes an *input* quantity and transforms it into an *output* quantity that can be observed or recorded, such as the movement of a pointer on a dial or the magnitude of a digital display. This chapter discusses the characteristics of both the input signals to a measurement system and the resulting output signals. The shape and form of a signal are often referred to as its *waveform*. The waveform contains information about the *magnitude* and *amplitude*, which indicate the size of the input quantity, and the *frequency*, which indicates the way the signal changes in time. An understanding of waveforms is required for the selection of measurement systems and the interpretation of measured signals.

2.2 INPUT/OUTPUT SIGNAL CONCEPTS

Two important tasks that engineers face in the measurement of physical variables are (1) selecting a measurement system and (2) interpreting the output from a measurement system. A simple example of selecting a measurement system might be the selection of a tire gauge for measuring the air pressure in a bicycle tire or in a car tire, as shown in Figure 2.1. The gauge for the car tire would be required to indicate pressures up to 275 kPa (40 lb/in.^{2}), but the bicycle tire gauge would be required to indicate higher pressures, maybe up to 700 kPa (100 lb/in.^{2}). This idea of the *range* of an instrument, its lower to upper measurement limits, is fundamental to all measurement systems and demonstrates that some basic understanding of the nature of the input signal, in this case the magnitude, is necessary in evaluating or selecting a measurement system for a particular application.

A much more difficult task is the evaluation of the output of a measurement system when the time or spatial behavior of the input is not known. The pressure in a tire does not change while we are trying to measure it, but what if we wanted to measure pressure in a cylinder in an automobile engine? Would the tire gauge or another gauge based on its operating principle work? We know that the pressure in the cylinder varies with time. If our task was to select a measurement system to determine this time-varying pressure, information about the pressure variations in the cylinder would be necessary. From thermodynamics and the speed range of the engine, it may be possible to estimate the magnitude of pressures to be expected and the rate with which they change. From that

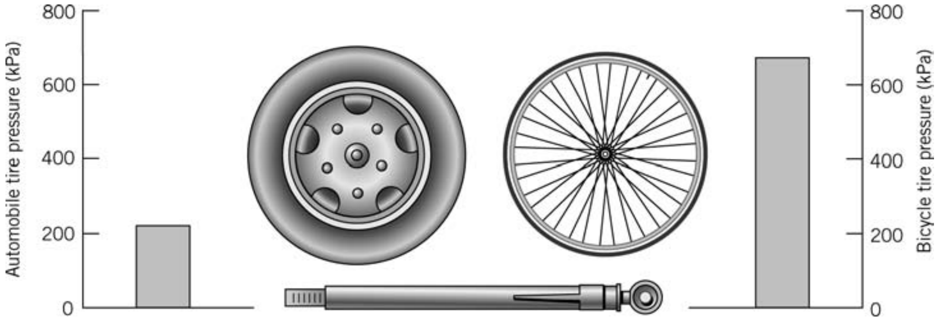


Figure 2.1 Measurement system selection based on input signal range.

we can select an appropriate measurement system. But to do this we need to develop a way to express this idea of the magnitude and rate of change of a variable.

Upon completion of this chapter, the reader will be able to

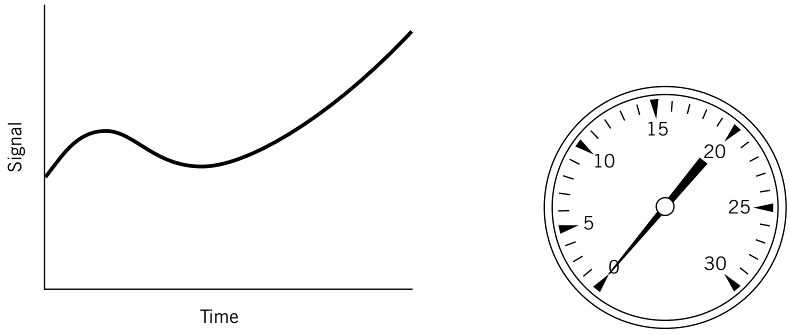
- define the range of an instrument,
- classify signals as analog, discrete time, or digital,
- compute the mean and rms values for time-varying signals, and
- characterize signals in the frequency domain.

Generalized Behavior

Many measurement systems exhibit similar responses under a variety of conditions, which suggests that the performance and capabilities of measuring systems may be described in a generalized way. To examine further the generalized behavior of measurement systems, we first examine the possible forms of the input and output signals. We will associate the term ‘‘signal’’ with the ‘‘transmission of information.’’ A *signal* is the physical information about a measured variable being transmitted between a process and the measurement system, between the stages of a measurement system, or as the output from a measurement system.

Classification of Waveforms

Signals may be classified as analog, discrete time, or digital. *Analog* describes a signal that is continuous in time. Because physical variables tend to be continuous, an analog signal provides a ready representation of their time-dependent behavior. In addition, the magnitude of the signal is continuous and thus can have any value within the operating range. An analog signal is shown in Figure 2.2a; a similar continuous signal would result from a recording of the pointer rotation with time for the output display shown in Figure 2.2b. Contrast this continuous signal with the signal shown in Figure 2.3a. This format represents a *discrete time signal*, for which information about the magnitude of the signal is available only at discrete points in time. A discrete time signal usually results from the sampling of a continuous variable at repeated finite time intervals. Because information in the signal shown in Figure 2.3a is available only at discrete times, some assumption must be made about the behavior of the measured variable during the times when it is not available. One approach is to assume the signal is constant between samples, a sample and hold method. The



(a) Analog signal representation **(b)** Analog display

Figure 2.2 Analog signal concepts.

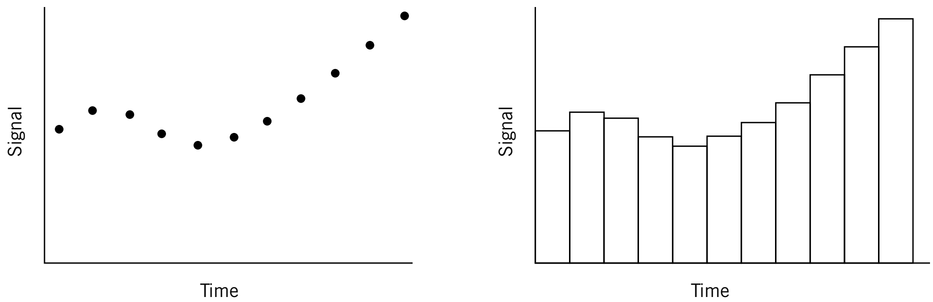
resulting waveform is shown in Figure 2.3b. Clearly, as the time between samples is reduced, the difference between the discrete variable and the continuous signal it represents decreases.

Digital signals are particularly useful when data acquisition and processing are performed using a digital computer. A digital signal has two important characteristics. First, a digital signal exists at discrete values in time, like a discrete time signal. Second, the magnitude of a digital signal is discrete, determined by a process known as quantization at each discrete point in time. *Quantization* assigns a single number to represent a range of magnitudes of a continuous signal.

Figure 2.4a shows digital and analog forms of the same signal where the magnitude of the digital signal can have only certain discrete values. Thus, a digital signal provides a quantized magnitude at discrete times. The waveform that would result from assuming that the signal is constant between sampled points in time is shown in Figure 2.4b.

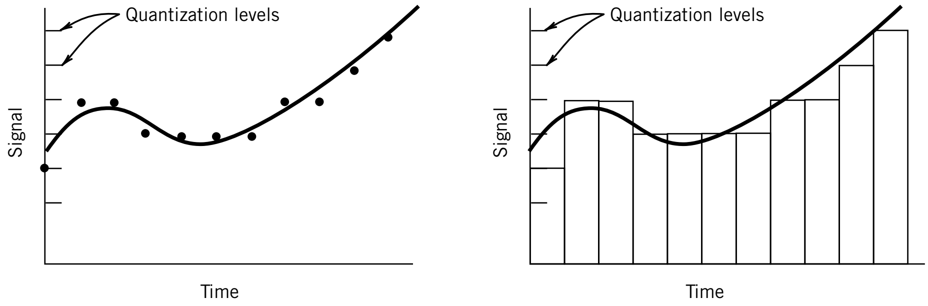
As an example of quantization, consider a digital clock that displays time in hours and minutes. For the entire duration of 1 minute, a single numerical value is displayed until it is updated at the next discrete time step. As such, the continuous physical variable of time is quantized in its conversion to a digital display.

Sampling of an analog signal to produce a digital signal can be accomplished by using an *analog-to-digital (A/D) converter*, a solid-state device that converts an analog voltage signal to a



(a) Discrete time signal **(b)** Discrete time waveform

Figure 2.3 Discrete time signal concepts.



(a) Digital signal

(b) Digital waveform

Figure 2.4 Digital signal representation and waveform.

binary number system representation. The limited resolution of the binary number that corresponds to a range of voltages creates the quantization levels and ranges.

For example, a compact disk player is built around technology that relies on the conversion of a continuously available signal, such as music from a microphone, into a digital form. The digital information is stored on a compact disk and later read in digital form by a laser playback system (1). However, to serve as input to a traditional stereo amplifier, and since speakers and the human ear are analog devices, the digital information is converted back into a continuous voltage signal for playback.

Signal Waveforms

In addition to classifying signals as analog, discrete time, or digital, some description of the waveform associated with a signal is useful. Signals may be characterized as either static or dynamic. A *static signal* does not vary with time. The diameter of a shaft is an example. Many physical variables change slowly enough in time, compared to the process with which they interact, that for all practical purposes these signals may be considered static in time. For example, the voltage across the terminals of a battery is approximately constant over its useful life. Or consider measuring temperature by using an outdoor thermometer; since the outdoor temperature does not change significantly in a matter of minutes, this input signal might be considered static when compared to our time period of interest. A mathematical representation of a static signal is given by a constant, as indicated in Table 2.1. In contrast, often we are interested in how the measured variable changes with time. This leads us to consider time-varying signals further.

A *dynamic signal* is defined as a time-dependent signal. In general, dynamic signal waveforms, $y(t)$, may be classified as shown in Table 2.1. A *deterministic signal* varies in time in a predictable manner, such as a sine wave, a step function, or a ramp function, as shown in Figure 2.5. A signal is *steady periodic* if the variation of the magnitude of the signal repeats at regular intervals in time. Examples of steady periodic behaviors are the motion of an ideal pendulum, and the temperature variations in the cylinder of an internal combustion engine under steady operating conditions. Periodic waveforms may be classified as simple or complex. A *simple periodic waveform* contains only one frequency. A *complex periodic waveform* contains multiple frequencies and is represented as a superposition of multiple simple periodic waveforms. *Aperiodic* is the term used to describe deterministic signals that do not repeat at regular intervals, such as a step function.

Table 2.1 Classification of Waveforms

I. Static	$y(t) = A_0$
II. Dynamic	
Periodic waveforms	
Simple periodic waveform	$y(t) = A_0 + C\sin(\omega t + \phi)$
Complex periodic waveform	$y(t) = A_0 + \sum_{n=1}^{\infty} C_n \sin(n\omega t + \phi_n)$
Aperiodic waveforms	
Step[a]	$y(t) = A_0 U(t)$ $= A_0$ for $t > 0$
Ramp	$y(t) = A_0 t$ for $0 < t < t_f$
Pulse[b]	$y(t) = A_0 U(t) - A_0 U(t - t_1)$
III. Nondeterministic waveform	$y(t) \approx A_0 + \sum_{n=1}^{\infty} C_n \sin(n\omega t + \phi_n)$

[a] $U(t)$ represents the unit step function, which is zero for $t < 0$ and 1 for $t \geq 0$.
[b] t_1 represents the pulse width.

Also described in Figure 2.5 is a *nondeterministic signal* that has no discernible pattern of repetition. A nondeterministic signal cannot be prescribed before it occurs, although certain characteristics of the signal may be known in advance. As an example, consider the transmission of data files from one computer to another. Signal characteristics such as the rate of data transmission and the possible range of signal magnitude are known for any signal in this system. However, it would not be possible to predict future signal characteristics based on existing information in such a signal. Such a signal is properly characterized as nondeterministic. Nondeterministic signals are generally described by their statistical characteristics or a model signal that represents the statistics of the actual signal.

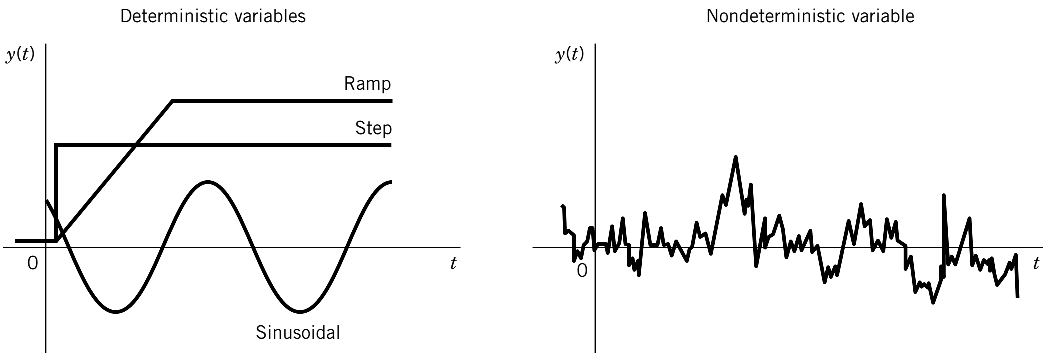


Figure 2.5 Examples of dynamic signals.

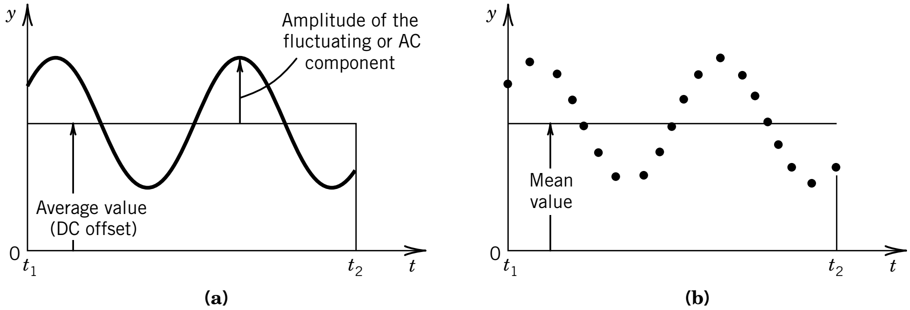


Figure 2.6 Analog and discrete representations of a dynamic signal.

2.3 SIGNAL ANALYSIS

In this section, we consider concepts related to the characterization of signals. A measurement system produces a signal that may be analog, discrete time, or digital. An analog signal is continuous with time and has a magnitude that is analogous to the magnitude of the physical variable being measured. Consider the analog signal shown in Figure 2.6a, which is continuous over the recorded time period from t_1 to t_2. The average or mean value[1] of this signal is found by

$$\bar{y} \equiv \frac{\int_{t_1}^{t_2} y(t)dt}{\int_{t_1}^{t_2} dt} \tag{2.1}$$

The mean value, as defined in Equation 2.1, provides a measure of the static portion of a signal over the time $t_2 - t_1$. It is sometimes called the *DC component* or DC offset of the signal.

The mean value does not provide any indication of the amount of variation in the dynamic portion of the signal. The characterization of the dynamic portion, or *AC component*, of the signal may be illustrated by considering the average power dissipated in an electrical resistor through which a fluctuating current flows. The power dissipated in a resistor due to the flow of a current is

$$P = I^2R$$

where

P = power dissipated = time rate of energy dissipated
I = current
R = resistance

If the current varies in time, the total electrical energy dissipated in the resistor over the time t_1 to t_2 would be

$$\int_{t_1}^{t_2} Pdt = \int_{t_1}^{t_2} [I(t)]^2 Rdt \tag{2.2}$$

The current $I(t)$ would, in general, include both a DC component and a changing AC component.

[1] Strictly speaking, for a continuous signal the mean value and the average value are the same. This is not true for discrete time signals.

Signal Root-Mean-Square Value

Consider finding the magnitude of a constant effective current, I_e, that would produce the same total energy dissipation in the resistor as the time-varying current, $I(t)$, over the time period t_1 to t_2. Assuming that the resistance, R, is constant, this current would be determined by equating $(I_e)^2R(t_2 - t_1)$ with Equation 2.2 to yield

$$I_e = \sqrt{\frac{1}{t_2 - t_1}\int_{t_1}^{t_2} [I(t)]^2\, dt} \tag{2.3}$$

This value of the current is called the root-mean-square (rms) value of the current. Based on this reasoning, the rms value of any continuous analog variable $y(t)$ over the time, $t_2 - t_1$, is expressed as

$$y_{\mathrm{rms}} = \sqrt{\frac{1}{t_2 - t_1}\int_{t_1}^{t_2} y^2 dt} \tag{2.4}$$

Discrete-Time or Digital Signals

A time-dependent analog signal, $y(t)$, can be represented by a discrete set of N numbers over the time period from t_1 to t_2 through the conversion

$$y(t) \rightarrow \{y(r\delta t)\} \quad r = 0, 1, \ldots, (N-1)$$

which uses the sampling convolution

$$\{y(r\delta t)\} = y(t)\delta(t - r\delta t) = \{y_i\} \quad i = 0, 1, 2, \ldots, (N-1)$$

Here, $\delta(t - r\delta t)$ is the delayed unit impulse function, δt is the sample time increment between each number, and $N\delta t = t_2 - t_1$ gives the total sample period over which the measurement of $y(t)$ takes place. The effect of discrete sampling on the original analog signal is demonstrated in Figure 2.6b in which the analog signal has been replaced by $\{y(r\delta t)\}$, which represents N values of a discrete time signal representing $y(t)$.

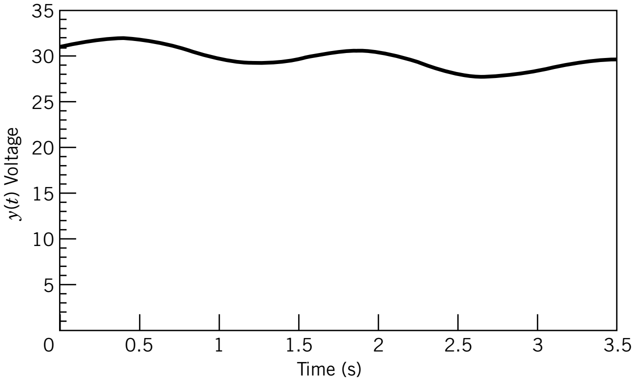
For either a discrete time signal or a digital signal, the mean value can be estimated by the discrete equivalent of Equation 2.1 as

$$\bar{y} = \frac{1}{N}\sum_{i=0}^{N-1} y_i \tag{2.5}$$

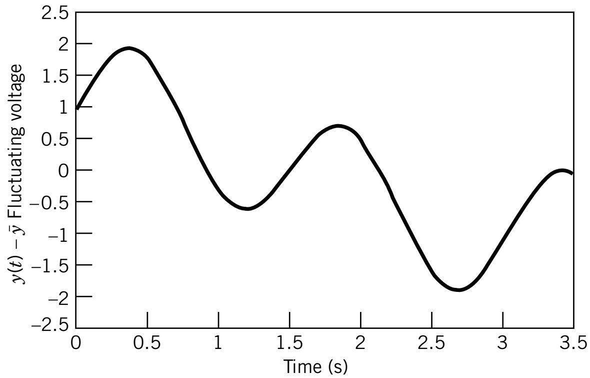
where each y_i is a discrete number in the data set of $\{y(r\delta t)\}$. The mean approximates the static component of the signal over the time interval t_1 to t_2. The rms value can be estimated by the discrete equivalent of Equation 2.4 as

$$y_{\mathrm{rms}} = \sqrt{\frac{1}{N}\sum_{i=0}^{N-1} y_i^2} \tag{2.6}$$

The rms value takes on additional physical significance when either the signal contains no DC component or the DC component has been subtracted from the signal. The rms value of a signal having a zero mean is a statistical measure of the magnitude of the fluctuations in the signal.



(a) Signal prior to subtracting DC offset



(b) Fluctuating component of signal

Figure 2.7 Effect of subtracting DC offset for a dynamic signal.

Direct Current Offset

When the AC component of the signal is of primary interest, the DC component can be removed. This procedure often allows a change of scale in the display or plotting of a signal such that fluctuations that were a small percentage of the DC signal can be more clearly observed without the superposition of the large DC component. The enhancement of fluctuations through the subtraction of the average value is illustrated in Figure 2.7.

Figure 2.7a contains the entire complex waveform consisting of both static (DC) and dynamic (AC) parts. When the DC component is removed, the characteristics of the fluctuating component of the signal are readily observed, as shown in Figure 2.7b. The statistics of the fluctuating component of the signal dynamic portion may contain important information for a particular application.

Example 2.1

Suppose the current passing through a resistor can be described by

$$I(t) = 10 \sin t$$

where I represents the time-dependent current in amperes. Establish the mean and rms values of current over a time from 0 to t_f, with $t_f = \pi$ and then with $t_f = 2\pi$. How do the results relate to the power dissipated in the resistor?

KNOWN $I(t) = 10 \sin t$

FIND $\bar{I}$ and I_{rms} with $t_f = \pi$ and 2π

SOLUTION The average value for a time from 0 to t_f is found from Equation 2.1 as

$$\bar{I} = \frac{\int_0^{t_f} I(t)dt}{\int_0^{t_f} dt} = \frac{\int_0^{t_f} 10 \sin t dt}{t_f}$$

Evaluation of this integral yields

$$\bar{I} = \frac{1}{t_f}[-10 \cos t]_0^{t_f}$$

With $t_f = \pi$, the average value, $\bar{I}$, is $20/\pi$. For $t_f = 2\pi$, the evaluation of the integral yields an average value of zero.

The rms value for the time period 0 to t_f is given by the application of Equation 2.4, which yields

$$I_{\text{rms}} = \sqrt{\frac{1}{t_f}\int_0^{t_f} I(t)^2 dt} = \sqrt{\frac{1}{t_f}\int_0^{t_f} (10 \sin t)^2 dt}$$

This integral is evaluated as

$$I_{\text{rms}} = \sqrt{\frac{100}{t_f}\left(-\frac{1}{2}\cos t \sin t + \frac{t}{2}\right)\Bigg|_0^{t_f}}$$

For $t_f = \pi$, the rms value is $\sqrt{50}$. Evaluation of the integral for the rms value with $t_f = 2\pi$ also yields $\sqrt{50}$.

COMMENT Although the average value over the period 2π is zero, the power dissipated in the resistor must be the same over both the positive and negative half-cycles of the sine function. Thus, the rms value of current is the same for the time period of π and 2π and is indicative of the rate at which energy is dissipated.

2.4 SIGNAL AMPLITUDE AND FREQUENCY

A key factor in measurement system behavior is the nature of the input signal to the system. A means is needed to classify waveforms for both the input signal and the resulting output signal relative to their magnitude and frequency. It would be very helpful if the behavior of measurement systems could be defined in terms of their response to a limited number and type of input signals. This is, in fact, exactly the case. A very complex signal, even one that is nondeterministic in nature, can be approximated as an infinite series of sine and cosine functions, as suggested in Table 2.1. The method of expressing such a complex signal as a series of sines and cosines is called *Fourier analysis*.

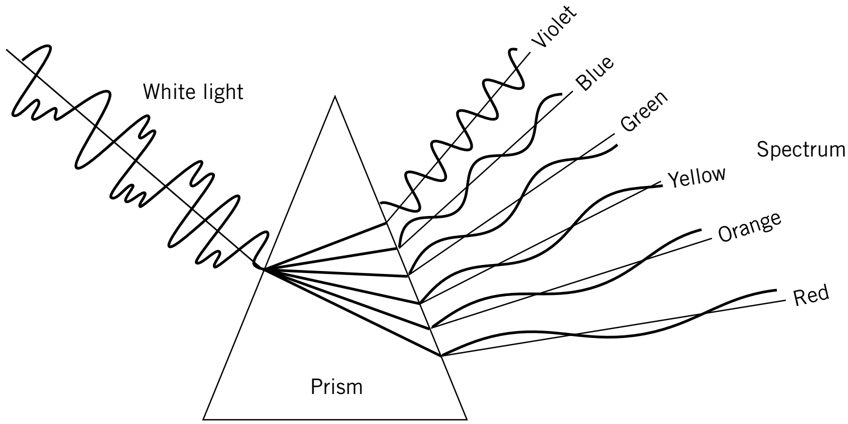


Figure 2.8 Separation of white light into its color spectrum. Color corresponds to a particular frequency or wavelength; light intensity corresponds to varying amplitudes.

Nature provides some experiences that support our contention that complex signals can be represented by the addition of a number of simpler periodic functions. For example, combining a number of different pure tones can generate rich musical sound. And an excellent physical analogy for Fourier analysis is provided by the separation of white light through a prism. Figure 2.8 illustrates the transformation of a complex waveform, represented by white light, into its simpler components, represented by the colors in the spectrum. In this example, the colors in the spectrum are represented as simple periodic functions that combine to form white light. Fourier analysis is roughly the mathematical equivalent of a prism and yields a representation of a complex signal in terms of simple periodic functions.

The representation of complex and nondeterministic waveforms by simple periodic functions allows measurement system response to be reasonably well defined by examining the output resulting from a few specific input waveforms, one of which is a simple periodic. As represented in Table 2.1, a simple periodic waveform has a single, well-defined amplitude and a single frequency. Before a generalized response of measurement systems can be determined, an understanding of the method of representing complex signals in terms of simpler functions is necessary.

Periodic Signals

The fundamental concepts of frequency and amplitude can be understood through the observation and analysis of periodic motions. Although sines and cosines are by definition geometric quantities related to the lengths of the sides of a right triangle, for our purposes sines and cosines are best thought of as mathematical functions that describe specific physical behaviors of systems. These behaviors are described by differential equations that have sines and cosines as their solutions. As an example, consider a mechanical vibration of a mass attached to a linear spring, as shown in Figure 2.9. For a linear spring, the spring force F and displacement y are related by $F = ky$, where k is the constant of proportionality, called the spring constant. Application of Newton's second law to this system yields a governing equation for the displacement y as a function of time t as

$$m\frac{d^2y}{dt^2} + ky = 0 \tag{2.7}$$

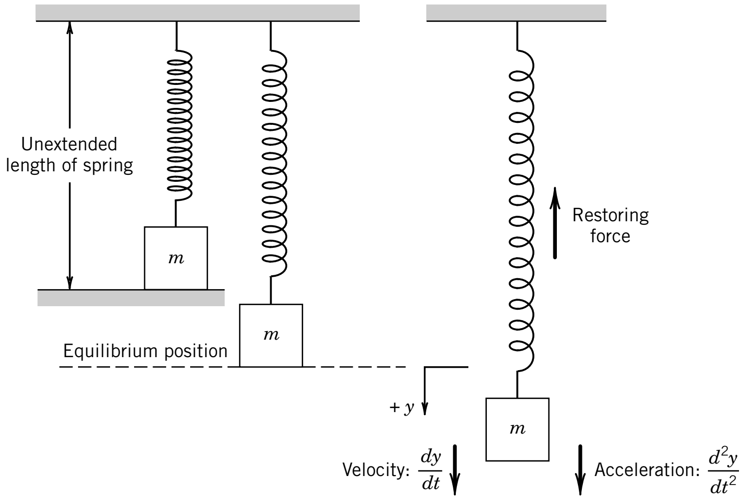


Figure 2.9 Spring-mass system.

This linear, second-order differential equation with constant coefficients describes the motion of the idealized spring mass system when there is no external force applied. The general form of the solution to this equation is

$$y = A\cos\omega t + B\sin\omega t \tag{2.8}$$

where $\omega = \sqrt{k/m}$. Physically we know that if the mass is displaced from the equilibrium point and released, it will oscillate about the equilibrium point. The time required for the mass to finish one complete cycle of the motion is called the *period*, and is generally represented by the symbol T.

Frequency is related to the period and is defined as the number of complete cycles of the motion per unit time. This *frequency*, f, is measured in cycles per second (Hz; 1 cycle/s = 1 Hz). The term ω is also a frequency, but instead of having units of cycles per second it has units of radians per second. This frequency, ω, is called the *circular frequency* since it relates directly to cycles on the unit circle, as illustrated in Figure 2.10. The relationship among ω, f, and the period, T, is

$$T = \frac{2\pi}{\omega} = \frac{1}{f} \tag{2.9}$$

In Equation 2.8, the sine and cosine terms can be combined if a phase angle is introduced such that

$$y = C\cos(\omega t - \phi) \tag{2.10a}$$

or

$$y = C\sin(\omega t + \phi^*) \tag{2.10b}$$

The values of C, ϕ, and ϕ^* are found from the following trigonometric identities:

$$\begin{aligned}
&A\cos\omega t + B\sin\omega t = \sqrt{A^2 + B^2}\cos(\omega t - \phi)\\
&A\cos\omega t + B\sin\omega t = \sqrt{A^2 + B^2}\sin(\omega t + \phi^*)\\
&\phi = \tan^{-1}\frac{B}{A} \quad \phi^* = \tan^{-1}\frac{A}{B} \quad \phi^* = \frac{\pi}{2} - \phi
\end{aligned} \tag{2.11}$$

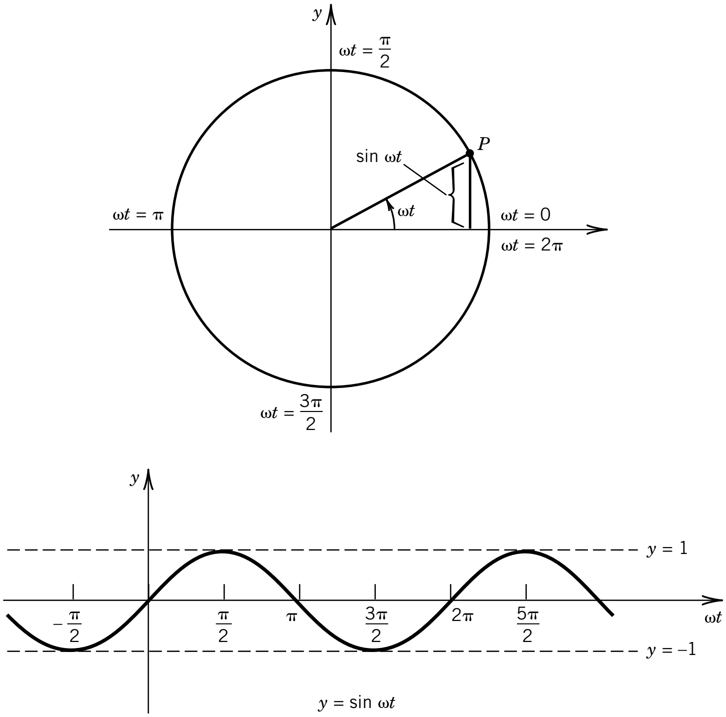


Figure 2.10 Relationship between cycles on the unit circle and circular frequency.

The size of the maximum and minimum displacements from the equilibrium position, or the value C, is the amplitude of the oscillation. The concepts of amplitude and frequency are essential for the description of time-dependent signals.

Frequency Analysis

Many signals that result from the measurement of dynamic variables are nondeterministic in nature and have a continuously varying rate of change. These signals, having *complex waveforms*, present difficulties in the selection of a measurement system and in the interpretation of an output signal. However, it is possible to separate a complex signal, or any signal for that matter, into a number of sine and cosine functions. *In other words, any complex signal can be thought of as made up of sines and cosines of differing periods and amplitudes, which are added together in an infinite trigonometric series.* This representation of a signal as a series of sines and cosines is called a *Fourier series*. Once a signal is broken down into a series of periodic functions, the importance of each frequency can be easily determined. This information about frequency content allows proper choice of a measurement system, and precise interpretation of output signals.

In theory, Fourier analysis allows essentially all mathematical functions of practical interest to be represented by an infinite series of sines and cosines.[2]

[2] A periodic function may be represented as a Fourier series if the function is piecewise continuous over the limits of integration and the function has a left- and right-hand derivative at each point in the interval. The sum of the resulting series is equal to the function at each point in the interval except points where the function is discontinuous.

The following definitions relate to Fourier analysis:

1. A function $y(t)$ is *a periodic function* if there is some positive number T such that

$$y(t+T) = y(t)$$

The *period* of $y(t)$ is T. If both $y_1(t)$ and $y_2(t)$ have period T, then

$$ay_1(t) + by_2(t)$$

also has a period of T (a and b are constants).

2. A *trigonometric series* is given by

$$A_0 + A_1 \cos t + B_1 \sin t + A_2 \cos 2t + B_2 \sin 2t + \ldots$$

where A_n and B_n are the coefficients of the series.

Example 2.2

As a physical (instead of mathematical) example of frequency content of a signal, consider stringed musical instruments, such as guitars and violins. When a string is caused to vibrate by plucking or bowing, the sound is a result of the motion of the string and the resonance of the instrument. (The concept of resonance is explored in Chapter 3.) The musical pitch for such an instrument is the lowest frequency of the string vibrations. Our ability to recognize differences in musical instruments is primarily a result of the higher frequencies present in the sound, which are usually integer multiples of the fundamental frequency. These higher frequencies are called *harmonics*.

The motion of a vibrating string is best thought of as composed of several basic motions that together create a musical tone. Figure 2.11 illustrates the vibration modes associated with a string plucked at its center. The string vibrates with a fundamental frequency and odd-numbered harmonics, each having a specific phase relationship with the fundamental. The relative strength of each harmonic is graphically illustrated through its amplitude in Figure 2.11. Figure 2.12 shows the motion, which is caused by plucking a string one-fifth of the distance from a fixed end. The resulting frequencies are illustrated in Figure 2.13. Notice that the fifth harmonic is missing from the resulting sound.

Musical sound from a vibrating string is analogous to a measurement system input or output, which contains many frequency components. Fourier analysis and frequency spectra provide insightful and practical means of reducing such complex signals into a combination of simple waveforms. Next, we explore the frequency and amplitude analysis of complex signals.

Available on the companion software site, *Program Sound.vi* uses your computer's microphone and sound board to sample ambient sounds and to decompose them into harmonics (try humming a tune).

Fourier Series and Coefficients

A periodic function $y(t)$ with a period $T = 2\pi$ is to be represented by a trigonometric series, such that for any t,

$$y(t) = A_0 + \sum_{n=1}^{\infty} (A_n \cos nt + B_n \sin nt) \tag{2.12}$$

With $y(t)$ known, the coefficients A_n and B_n are to be determined; this requires a well-established mathematical procedure, but not one that we need to reinvent. For A_0 to be determined,

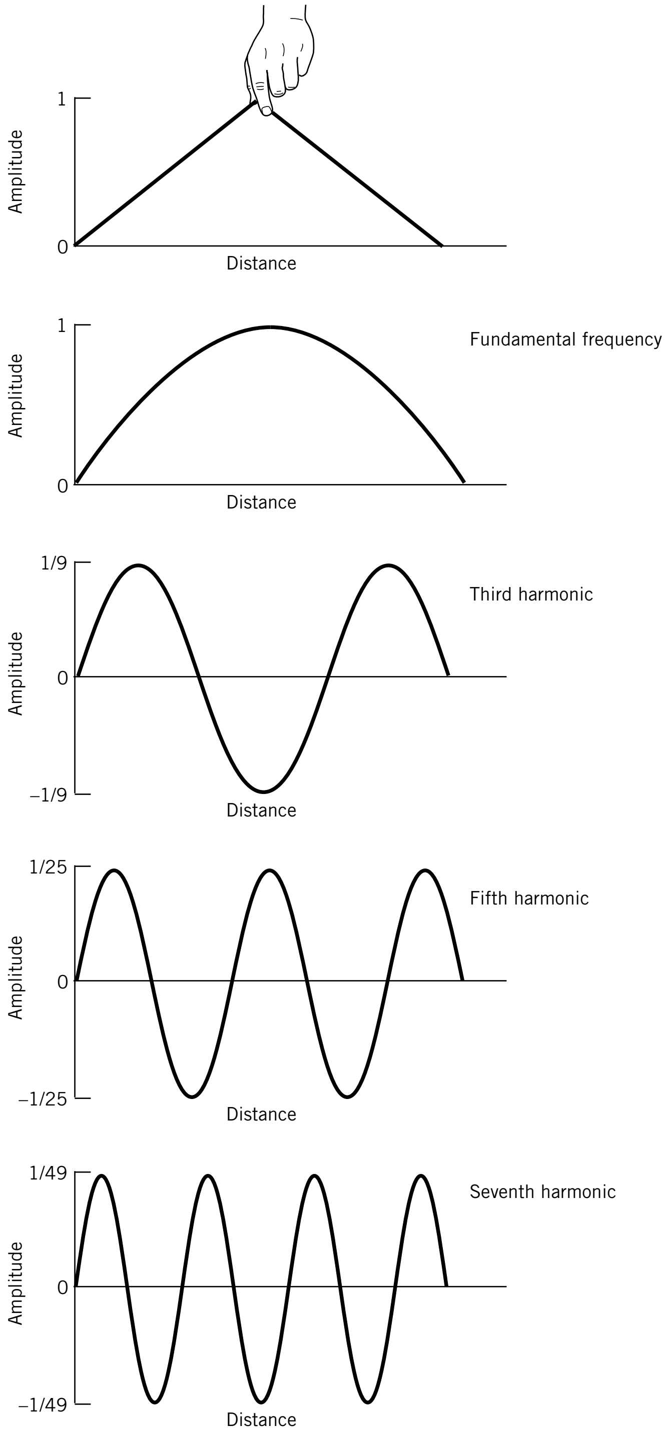


Figure 2.11 Modes of vibration for a string plucked at its center.

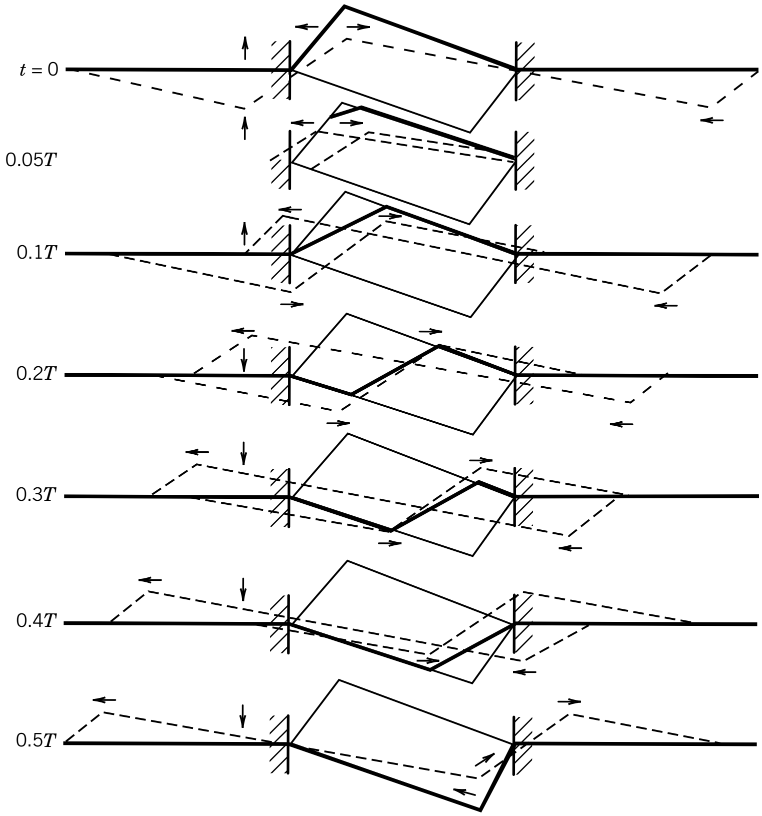


Figure 2.12 Motion of a string plucked one-fifth of the distance from a fixed end. (From N. H. Fletcher and T. D. Rossing, *The Physics of Musical Instruments*. Copyright © 1991 by Springer-Verlag, New York. Reprinted by permission.)

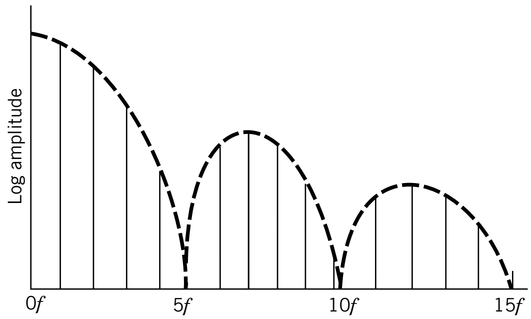


Figure 2.13 Frequency spectrum for a string plucked one-fifth of the distance from a fixed end. (From N. H. Fletcher and T. D. Rossing, *The Physics of Musical Instruments*. Copyright © 1991 by Springer-Verlag, New York. Reprinted by permission.)

Equation 2.12 is integrated from $-\pi$ to π:

$$\int_{-\pi}^{\pi} y(t)dt = A_0 \int_{-\pi}^{\pi} dt + \sum_{n=1}^{\infty} \left(A_n \int_{-\pi}^{\pi} \cos ntdt + B_n \int_{-\pi}^{\pi} \sin ntdt \right) \tag{2.13}$$

Since

$$\int_{-\pi}^{\pi} \cos ntdt = 0 \quad \text{and} \quad \int_{-\pi}^{\pi} \sin ntdt = 0$$

Equation 2.13 yields

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} y(t)dt \tag{2.14}$$

The coefficient A_m may be determined by multiplying Equation 2.12 by cos mt and integrating from $-\pi$ to π. The resulting expression for A_m is

$$A_m = \frac{1}{\pi} \int_{-\pi}^{\pi} y(t)\cos mtdt \tag{2.15}$$

Similarly, multiplying Equation 2.12 by sin mt and integrating from $-\pi$ to π yields B_m. Thus, for a function $y(t)$ with a period 2π, the coefficients of the trigonometric series representing $y(t)$ are given by the Euler formulas:

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} y(t) \\ A_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} y(t)\cos ntdt \\ B_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} y(t)\sin ntdt \end{aligned} \tag{2.16}$$

The trigonometric series corresponding to $y(t)$ is called the *Fourier series for* $y(t)$, and the coefficients A_n and B_n are called the *Fourier coefficients* of $y(t)$. In the series for $y(t)$ in Equation 2.12, when $n = 1$ the corresponding terms in the Fourier series are called *fundamental* and have the lowest frequency in the series. The *fundamental frequency* for this Fourier series is $\omega = 2\pi/2\pi = 1$. Frequencies corresponding to $n = 2, 3, 4, \ldots$ are known as *harmonics*, with, for example, $n = 2$ representing the second harmonic.

Functions represented by a Fourier series normally do not have a period of 2π. However, the transition to an arbitrary period can be affected by a change of scale, yielding a new set of Euler formulas described below.

Fourier Coefficients for Functions Having Arbitrary Periods

The coefficients of a trigonometric series representing a function having an arbitrary period T are given by the Euler formulas:

$$
\begin{aligned}
A_0 &= \frac{1}{T}\int_{-T/2}^{T/2} y(t)dt \\
A_n &= \frac{2}{T}\int_{-T/2}^{T/2} y(t)\cos n\omega t dt \\
B_n &= \frac{2}{T}\int_{-T/2}^{T/2} y(t)\sin n\omega t dt
\end{aligned}
\tag{2.17}
$$

where $n = 1, 2, 3, \ldots$, and $T = 2\pi/\omega$ is the period of $y(t)$. The trigonometric series that results from these coefficients is a Fourier series and may be written as

$$y(t) = A_0 + \sum_{n=1}^{\infty} (A_n \cos n\omega t + B_n \sin n\omega t) \tag{2.18}$$

A series of sines and cosines may be written as a series of either sines or cosines through the introduction of a phase angle, so that the Fourier series in Equation 2.18,

$$y(t) = A_0 + \sum_{n=1}^{\infty} (A_n \cos n\omega t + B_n \sin n\omega t)$$

may be written as

$$y(t) = A_0 + \sum_{n=1}^{\infty} C_n \cos (n\omega t - \phi_n) \tag{2.19}$$

or

$$y(t) = A_0 + \sum_{n=1}^{\infty} C_n \sin \left(n\omega t + \phi_n^*\right) \tag{2.20}$$

where

$$
\begin{aligned}
C_n &= \sqrt{A_n^2 + B_n^2} \\
\tan \phi_n &= \frac{B_n}{A_n} \quad \text{and} \quad \tan \phi_n^* = \frac{A_n}{B_n}
\end{aligned}
\tag{2.21}
$$

Even and Odd Functions

A function $g(t)$ is even if it is symmetric about the origin, which may be stated, for all t,

$$g(-t) = g(t)$$

A function $h(t)$ is odd if, for all t,

$$h(-t) = -h(t)$$

For example, $\cos nt$ is even, while $\sin nt$ is odd. A particular function or waveform may be even, odd, or neither even nor odd.

Fourier Cosine Series

If $y(t)$ is even, its Fourier series will contain only cosine terms:

$$y(t) = \sum_{n=1}^{\infty} A_n \cos \frac{2\pi nt}{T} = \sum_{n=1}^{\infty} A_n \cos n\omega t \tag{2.22}$$

Fourier Sine Series

If $y(t)$ is odd, its Fourier series will contain only sine terms

$$y(t) = \sum_{n=1}^{\infty} B_n \sin \frac{2\pi nt}{T} = \sum_{n=1}^{\infty} B_n \sin n\omega t \tag{2.23}$$

Note: Functions that are neither even nor odd result in Fourier series that contain both sine and cosine terms.

Example 2.3

Determine the Fourier series that represents the function shown in Figure 2.14.

KNOWN $T = 10$ (i.e., -5 to $+5$)

$A_0 = 0$

FIND The Fourier coefficients $A_1, A_2, \ldots$ and $B_1, B_2, \ldots$

SOLUTION Since the function shown in Figure 2.14 is odd, the Fourier series will contain only sine terms (see Eq. 2.23):

$$y(t) = \sum_{n=1}^{\infty} B_n \sin \frac{2\pi nt}{T}$$

where

$$B_n = \frac{2}{10}\left[\int_{-5}^{0} (-1)\sin\left(\frac{2\pi nt}{10}\right) dt + \int_{0}^{5} (1)\sin\left(\frac{2\pi nt}{10}\right) dt\right]$$

$$B_n = \frac{2}{10}\left\{\left[\frac{10}{2n\pi}\cos\left(\frac{2\pi nt}{10}\right)\right]_{-5}^{0} + \left[\frac{-10}{2n\pi}\cos\left(\frac{2\pi nt}{10}\right)\right]_{0}^{5}\right\}$$

$$B_n = \frac{2}{10}\left\{\frac{10}{2n\pi}\left[1 - \cos(-n\pi) - \cos(n\pi) + 1\right]\right\}$$

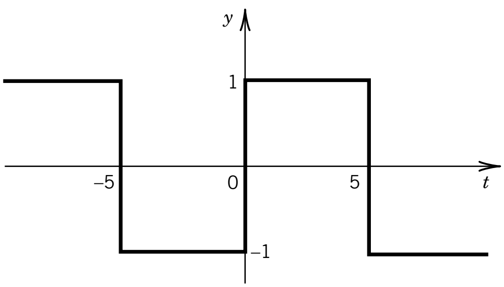


Figure 2.14 Function represented by a Fourier series in Example 2.3.

The reader can verify that all $A_n = 0$ for this function. For even values of n, B_n is identically zero and for odd values of n,

$$B_n = \frac{4}{n\pi}$$

The resulting Fourier series is then

$$y(t) = \frac{4}{\pi}\sin\frac{2\pi}{10}t + \frac{4}{3\pi}\sin\frac{6\pi}{10}t + \frac{4}{5\pi}\sin\frac{10\pi}{10}t + \cdots$$

Note that the fundamental frequency is $\omega = \frac{2\pi}{10}$ rad/s and the subsequent terms are the odd-numbered harmonics of ω.

COMMENT Consider the function given by

$$y(t) = 1 \quad 0 < t < 5$$

We may represent $y(t)$ by a Fourier series if we extend the function beyond the specified range (0–5) either as an even periodic extension or as an odd periodic extension. (Because we are interested only in the Fourier series representing the function over the range $0 < t < 5$, we can impose any behavior outside of that domain that helps to generate the Fourier coefficients!) Let's choose an odd periodic extension of $y(t)$; the resulting function remains identical to the function shown in Figure 2.14.

Example 2.4

Find the Fourier coefficients of the periodic function

$$y(t) = -5 \text{ when } -\pi < t < 0$$
$$y(t) = +5 \text{ when } 0 < t < \pi$$

and $y(t + 2\pi) = y(t)$. Plot the resulting first four partial sums for the Fourier series.

KNOWN Function $y(t)$ over the range $-\pi$ to π

FIND Coefficients A_n and B_n

SOLUTION The function as stated is periodic, having a period of $T = 2\pi$ (i.e., $\omega = 1$ rad/s), and is identical in form to the odd function examined in Example 2.3. Since this function is also odd, the Fourier series contains only sine terms, and

$$B_n = \frac{1}{\pi}\int_{-\pi}^{\pi} y(t)\sin n\omega t\,dt = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-5)\sin nt\,dt + \int_{0}^{\pi}(+5)\sin nt\,dt\right]$$

which yields upon integration

$$\frac{1}{\pi}\left\{\left[(5)\frac{\cos nt}{n}\right]_{-\pi}^{0} - \left[(5)\frac{\cos nt}{n}\right]_{0}^{\pi}\right\}$$

Thus,

$$B_n = \frac{10}{n\pi}(1 - \cos n\pi)$$

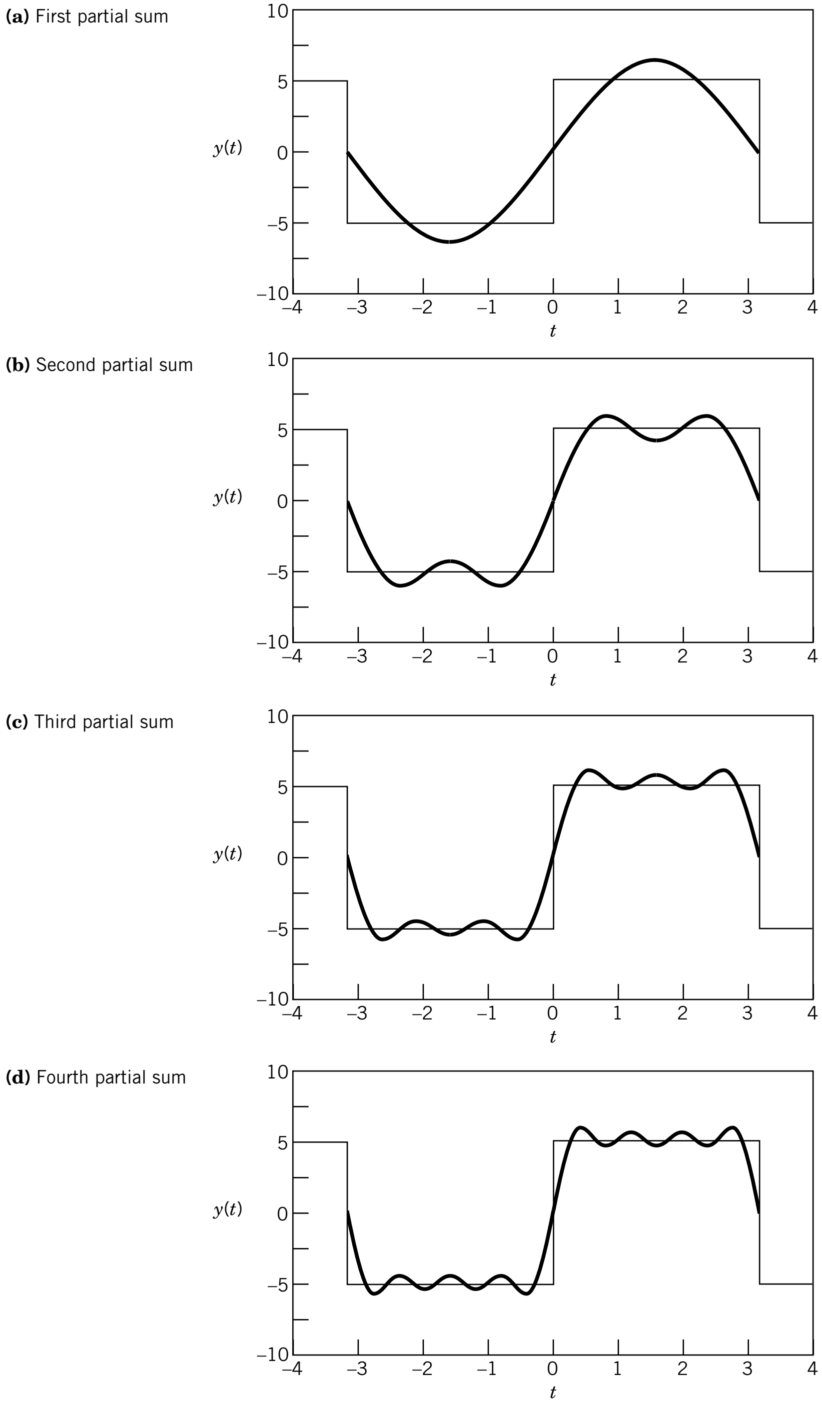


Figure 2.15 First four partial sums of the Fourier series $(20/\pi)(\sin t + 1/3 \sin 3t + 1/5 \sin 5t + \cdots)$ in comparison with the exact waveform.

which is zero for even n, and $20/n\pi$ for odd values of n. The Fourier series can then be written[3]

$$y(t) = \frac{20}{\pi}\left(\sin t + \frac{1}{3}\sin 3t + \frac{1}{5}\sin 5t + \cdots\right)$$

Figure 2.15 shows the first four partial sums of this function, as they compare with the function they represent. Note that at the points of discontinuity of the function, the Fourier series takes on the arithmetic mean of the two function values.

COMMENT As the number of terms in the partial sum increases, the Fourier series approximation of the square wave function becomes even better. For each additional term in the series, the number of "humps" in the approximate function in each half-cycle corresponds to the number of terms included in the partial sum.

The Matlab®[4] program file *FourCoef* with the companion software illustrates the behavior of the partial sums for several waveforms. The LabView® program *Waveform-Generation.vi* creates signals from trigonometric series.

Example 2.5

As an example of interpreting the frequency content of a given signal, consider the output voltage from a rectifier. A rectifier functions to "flip" the negative half of an alternating current (AC) into the positive half plane, resulting in a signal that appears as shown in Figure 2.16. For the AC signal the voltage is given by

$$E(t) = 120 \sin 120\pi t$$

The period of the signal is 1/60 s, and the frequency is 60 Hz.

KNOWN The rectified signal can be expressed as

$$E(t) = |120 \sin 120\pi t|$$

FIND The frequency content of this signal as determined from a Fourier series analysis.

SOLUTION The frequency content of this signal can be determined by expanding the function in a Fourier series. The coefficients may be determined using the Euler formulas, keeping in mind

---

[3] If we assume that the sum of this series most accurately represents the function y at $t = \pi/2$, then

$$y\left(\frac{\pi}{2}\right) = 5 = \frac{20}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - + \cdots\right)$$

or

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + - \cdots$$

This series approximation of π was first obtained by Gottfried Wilhelm Leibniz (1646–1716) in 1673 from geometrical reasoning.

[4] Matlab is a registered trademark of Mathworks, Inc.

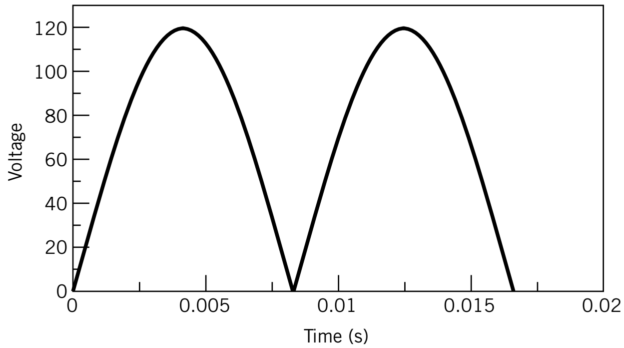


Figure 2.16 Rectified sine wave.

that the rectified sine wave is an even function. The coefficient A_0 is determined from

$$A_0 = 2\left[\frac{1}{T}\int_0^{T/2} y(t)dt\right] = \frac{2}{1/60}\int_0^{1/120} 120\sin 120\pi t dt$$

with the result that

$$A_0 = \frac{2 \times 60 \times 2}{\pi} = 76.4$$

The remaining coefficients in the Fourier series may be expressed as

$$\begin{aligned} A_n &= \frac{4}{T}\int_0^{T/2} y(t)\cos\frac{2n\pi t}{T} dt \\ &= \frac{4}{1/60}\int_0^{1/120} 120\sin 120\,\pi t\cos n\pi t dt \end{aligned}$$

For values of n that are odd, the coefficient A_n is identically zero. For values of n that are even, the result is

$$A_n = \frac{120}{\pi}\left(\frac{-2}{n-1} + \frac{2}{n+1}\right) \tag{2.24}$$

The Fourier series for the function $|120\sin 120\pi t|$ is

$$76.4 - 50.93\cos 240\pi t - 10.10\cos 480\pi t - 4.37\cos 720\pi t \ldots$$

Figure 2.17 shows the amplitude versus frequency content of the rectified signal, based on a Fourier series expansion.

COMMENT The frequency content of a signal is determined by examining the amplitude of the various frequency components that are present in the signal. For a periodic mathematical function, expanding the function in a Fourier series and plotting the amplitudes of the contributing sine and cosine terms can illustrate these frequency contributions.

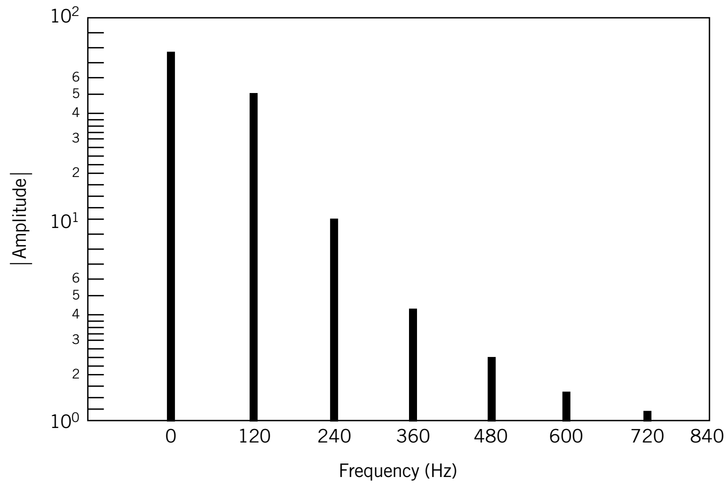


Figure 2.17 Frequency content of the function $y(t) = |120 \sin 20\pi t|$ displayed as an amplitude-frequency spectrum.

The LabView program *WaveformGeneration.vi* follows Examples 2.4 and 2.5 and provides the spectra for a number of signals. Several Matlab programs are provided also (e.g., *FourCoef*, *FunSpect*, *DataSpect*) to explore the concept of superposition of simple periodic signals to create a more complex signal. The software allows you to create your own functions and then explore their frequency and amplitude content.

2.5 FOURIER TRANSFORM AND THE FREQUENCY SPECTRUM

The previous discussion of Fourier analysis demonstrates that an arbitrary, but known, function can be expressed as a series of sines and cosines known as a Fourier series. The coefficients of the Fourier series specify the amplitudes of the sines and cosines, each having a specific frequency. Unfortunately, in most practical measurement applications the input signal may not be known in functional form. Therefore, although the theory of Fourier analysis demonstrates that any function can be expressed as a Fourier series, the analysis presented so far has not provided a specific technique for analyzing measured signals. Such a technique for the decomposition of a measured dynamic signal in terms of amplitude and frequency is now described.

Recall that the dynamic portion of a signal of arbitrary period can be described from Equation 2.17.

$$
\begin{aligned}
A_n &= \frac{2}{T}\int_{-T/2}^{T/2} y(t) \cos n\omega t dt \\
B_n &= \frac{2}{T}\int_{-T/2}^{T/2} y(t) \sin n\omega t dt
\end{aligned}
\tag{2.25}
$$

where the amplitudes A_n and B_n correspond to the nth frequency of a Fourier series.

If we consider the period of the function to approach infinity, we can eliminate the constraint on Fourier analysis that the signal be a periodic waveform. In the limit as T approaches infinity the Fourier series becomes an integral. The spacing between frequency components becomes infinitesimal. This means that the coefficients A_n and B_n become continuous functions of frequency and can be expressed as $A(\omega)$ and $B(\omega)$ where

$$\begin{aligned} A(\omega) &= \int_{-\infty}^{\infty} y(t)\cos\omega t dt \\ B(\omega) &= \int_{-\infty}^{\infty} y(t)\sin\omega t dt \end{aligned} \tag{2.26}$$

The Fourier coefficients $A(\omega)$ and $B(\omega)$ are known as the components of the Fourier transform of $y(t)$.

To develop the Fourier transform, consider the complex number defined as

$$Y(\omega) \equiv A(\omega) - iB(\omega) \tag{2.27}$$

where $i = \sqrt{-1}$. Then from Equations 2.26 it follows directly that

$$Y(\omega) \equiv \int_{-\infty}^{\infty} y(t)(\cos\omega t - i\sin\omega t)dt \tag{2.28}$$

Introducing the identity

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

leads to

$$Y(\omega) \equiv \int_{-\infty}^{\infty} y(t)e^{-i\omega t}dt \tag{2.29}$$

Alternately, recalling from Equation 2.9 that the cyclical frequency f, in hertz, is related to the circular frequency and its period by

$$f = \frac{\omega}{2\pi} = \frac{1}{T}$$

Equation 2.29 is rewritten as

$$Y(f) \equiv \int_{-\infty}^{\infty} y(t)e^{-i2\pi f t}dt \tag{2.30}$$

Equation 2.29 or 2.30 provides the two-sided *Fourier transform* of $y(t)$. If $y(t)$ is known, then its Fourier transform will provide the amplitude-frequency properties of the signal, $y(t)$, which otherwise are not readily apparent in its time-based form. We can think of the Fourier transform as a decomposition of $y(t)$ into amplitude versus frequency information. This property is analogous to the optical properties displayed by the prism in Figure 2.8.

If $Y(f)$ is known or measured, we can recover the signal $y(t)$ from

$$y(t) = \int_{-\infty}^{\infty} Y(f)e^{i2\pi f t}df \tag{2.31}$$

Equation 2.31 describes the *inverse Fourier transform* of $Y(f)$. It suggests that given the amplitude-frequency properties of a signal we can reconstruct the original signal $y(t)$. The Fourier transform is a complex number having a magnitude and a phase,

$$Y(f) = |Y(f)|e^{i\phi(f)} = A(f) - iB(f) \tag{2.32}$$

The magnitude of $Y(f)$, also called the modulus, is given by

$$|Y(f)| = \sqrt{Re[Y(f)]^2 + \mathrm{Im}[Y(f)]^2} \tag{2.33}$$

and the phase by

$$\phi(f) = \tan^{-1}\frac{\mathrm{Im}[Y(f)]}{\mathrm{Re}[Y(f)]} \tag{2.34}$$

As noted earlier, the Fourier coefficients are related to cosine and sine terms. Then the amplitude of $y(t)$ can be expressed by its amplitude-frequency spectrum, or simply referred to as its amplitude spectrum, by

$$C(f) = \sqrt{A(f)^2 + B(f)^2} \tag{2.35}$$

and its phase spectrum by

$$\phi(f) = \tan^{-1}\frac{B(f)}{A(f)} \tag{2.36}$$

Thus the introduction of the Fourier transform provides a method to decompose a measured signal $y(t)$ into its amplitude-frequency components. Later we will see how important this method is, particularly when digital sampling is used to measure and interpret an analog signal.

A variation of the amplitude spectrum is the power spectrum, which is given by magnitude $C(f)^2/2$. Further details concerning the properties of the Fourier transform and spectrum functions can be found in Bracewell (2) and Champeney (3). An excellent historical account and discussion of the wide-ranging applications are found in Bracewell (4).

Discrete Fourier Transform

As a practical matter, it is likely that if $y(t)$ is measured and recorded, then it will be stored in the form of a discrete time or digital signal. A computer-based data-acquisition system is the most common method for recording data. These data are acquired over a finite period of time rather than the mathematically convenient infinite period of time. A discrete data set containing N values representing a time interval from 0 to t_f will accurately represent the signal provided that the measuring period has been properly chosen and is sufficiently long. We deal with the details for such period selection in a discussion on sampling concepts in Chapter 7. The preceding analysis is now extended to accommodate a discrete series.

Consider the time-dependent portion of the signal $y(t)$, which is measured N times at equally spaced time intervals δt. In this case, the continuous signal $y(t)$ is replaced by the discrete time signal given by $y(r\delta t)$ for $r = 0, 1, \ldots, (N-1)$. In effect, the relationship between $y(t)$ and $\{y(r\delta t)\}$ is described by a set of impulses of an amplitude determined by the value of $y(t)$ at each time step $r\delta t$. This transformation from a continuous to discrete time signal is described by

$$\{y(r\delta t)\} = y(t)\delta(t - r\delta t) \quad r = 0, 1, 2, \ldots, N-1 \tag{2.37}$$

where $\delta(t - r\delta t)$ is the delayed unit impulse function and $\{y(r\delta t)\}$ refers to the discrete data set given by $y(r\delta t)$ for $r = 0, 1, 2, \ldots, N-1$.

An approximation to the Fourier transform integral of Equation 2.30 for use on a discrete data set is the discrete Fourier transform (DFT). The DFT is given by

$$
\begin{aligned}
Y(f_k) &= \frac{2}{N}\sum_{r=0}^{N-1} y(r\delta t)e^{-i2\pi rk/N} \\
f_k &= k\delta f \qquad\qquad k = 0,\ 1,\ 2, \ldots, \left(\frac{N}{2}-1\right) \\
\delta f &= 1/N\delta t
\end{aligned}
\tag{2.38}
$$

Here, δf is the frequency resolution of the DFT with each value of $Y(f_k)$ corresponding to frequency increments of δf. In developing Equation 2.38 from Equation 2.30, t was replaced by $r\delta t$ and f replaced by $k/N\delta t$. The factor $2/N$ scales the transform when it is obtained from a data set of finite length (use $1/N$ for $k=0$ only).

The DFT as expressed by Equation 2.38 yields $N/2$ discrete values of the Fourier transform of $\{y(r\delta t)\}$. This is the so-called one-sided or half-transform as it assumes that the data set is one-sided, extending from 0 to t_f, and it returns only positive valued frequencies.

Equation 2.38 performs the numerical integration required by the Fourier integral. Equations 2.35, 2.36, and 2.38 demonstrate that the application of the DFT on the discrete series of data, $y(r\delta t)$, permits the decomposition of the discrete data in terms of frequency and amplitude content. Hence, by using this method a measured discrete signal of unknown functional form can be reconstructed as a Fourier series through Fourier transform techniques.

Software for computing the Fourier transform of a discrete signal is included in the companion software. The time required to compute directly the DFT algorithm described in this section increases at a rate that is proportional to N^2. This makes it inefficient for use with data sets of large N. A fast algorithm for computing the DFT, known as the *fast Fourier transform* (FFT), was developed by Cooley and Tukey (5). This method is widely available and is the basis for most Fourier analysis software packages. The FFT algorithm is discussed in most advanced texts on signal analysis (6, 7). The accuracy of discrete Fourier analysis depends on the frequency content[5] of $y(t)$ and on the Fourier transform frequency resolution. An extensive discussion of these interrelated parameters is given in Chapter 7.

Example 2.6

Convert the continuous signal described by $y(t) = 10 \sin 2\pi t$ V into a discrete set of eight numbers using a time increment of 0.125 s.

KNOWN The signal has the form $y(t) = C_1 \sin 2\pi f_1 t$
where

$$
\begin{aligned}
f_1 &= \omega_1/2\pi = 1\ \text{Hz} \\
C(f_1 &= 1\ \text{Hz}) = 10\ \text{V} \\
\phi(f_1) &= 0 \\
\delta t &= 0.125\ \text{s} \\
N &= 8
\end{aligned}
$$

[5] The value of $1/\delta t$ must be more than twice the highest frequency contained in $y(t)$.

Table 2.2 Discrete Data Set for $y(t) = 10 \sin 2\pi t$

r	$y(r\delta t)$	r	$y(r\delta t)$
0	0.000	4	0.000
1	7.071	5	−7.071
2	10.000	6	−10.000
3	7.071	7	−7.071

FIND $\{y(r\delta t)\}$

SOLUTION Measuring $y(t)$ every 0.125 s over 1 s produces the discrete data set $\{y(r\delta t)\}$ given in Table 2.2. Note that the measurement produces four complete periods of the signal and that the signal duration is given by $N\delta t = 1$ s. The signal and its discrete representation as a series of impulses in a time domain are plotted in Figure 2.18. See Example 2.7 for more on how to create this discrete series by using common software.

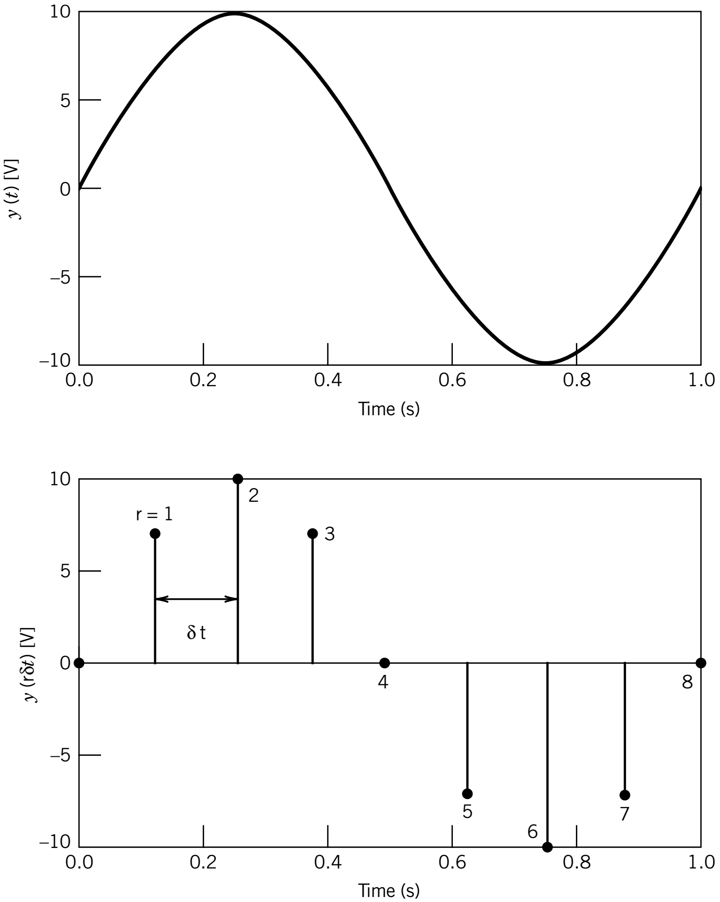


Figure 2.18 Representation of a simple periodic function as a discrete signal.

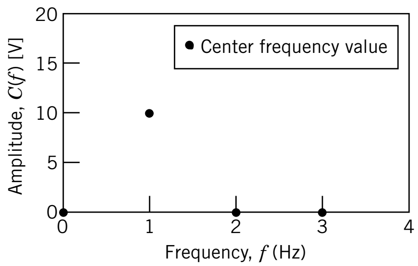


Figure 2.19 Amplitude as a function of frequency for a discrete representation of 10 sin 2πt resulting from a discrete Fourier transform algorithm.

Example 2.7

Estimate the amplitude spectrum of the discrete data set in Example 2.6.

KNOWN Discrete data set $\{y(r\delta t)\}$ of Table 2.2

$\delta t = 0.125$ s

$N = 8$

FIND $C(f)$

SOLUTION We can use either the Matlab Fourier transform software (file *FunSpect*), the LabView program *WaveformGeneration*, or a spreadsheet program to find $C(f)$ (see the Comment for details). For $N = 8$, a one-sided Fourier transform algorithm will return $N/2$ values for $C(f)$, with each successive amplitude corresponding to a frequency spaced at intervals of $1/N\delta t = 0.125$ Hz. The amplitude spectrum is shown in Figure 2.19 and has a spike of 10 V centered at a frequency of 1 Hz.

COMMENT The discrete series and the amplitude spectrum are easily reproduced by using the Matlab program file called *FunSpect*, the LabView program *Waveform-Generation*, or spreadsheet software. The Fourier analysis capability of this software can also be used on an existing data set, such as with the Matlab program file called *DataSpect*.

The following discussion of Fourier analysis using a spreadsheet program makes specific references to procedures and commands from Microsoft® Excel;[6] similar functions are available in other engineering analysis software. When a spreadsheet is used as shown, the $N = 8$ data point sequence $\{y(r\delta t)\}$ is created as in column 3. Under Data/Data Analysis, select Fourier Analysis. At the prompt, define the N cells containing $\{y(r\delta t)\}$ and define the cell destination (column 4). The analysis executes the DFT of Equation 2.38 by using the FFT algorithm, and it returns N complex numbers, the Fourier coefficients $Y(f)$ of Equation 2.32. The analysis returns the two-sided transform. However, a finite data set is one-sided, so we are interested in only the first $N/2$ Fourier coefficients of column 4 (the second $N/2$ coefficients just mirror the first $N/2$). To find the coefficient magnitude as given in Equation 2.33, compute or use the function IMABS$(= \sqrt{A^2 + B^2})$ on each of the first $N/2$ coefficients, and then scale each coefficient magnitude by dividing by $N/2$ (for $r = 0$,

[6] Microsoft® and Excel are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries.

divide by N). The $N/2$ scaled coefficients (column 5) now represent the discrete amplitudes corresponding to $N/2$ discrete frequencies (column 6) extending from $f = 0$ to $(N/2 - 1)/N\delta t$ Hz, with each frequency separated by $1/N\delta t$.

Column					
1	2	3	4	5	6
r	$t(s)$	$y(r\delta t)$	$Y(f)$=A - Bi	$C(f)$	f(Hz)
0	0	0	0	0	0
1	0.125	7.07	−40i	10	1
2	0.25	10	0	0	2
3	0.375	7.07	0	0	3
4	0.5	0	0		
5	0.625	−7.07	0		
6	0.75	−10	0		
7	0.875	−7.07	40i		

These same operations in Matlab are as follows:

$t = 1/8$: 1/8: 1	defines time from 0.125 s to 1 s in increments of 0.125 s
$y = 10*\sin(2*\text{pi}*t)$	creates the discrete time series with $N = 8$
ycoef = fft(y)	performs the Fourier analysis; returns N coefficients
c = coef/4	divides by $N/2$ to scale and determine the magnitudes

When signal frequency content is not known prior to conversion to a discrete signal, it is necessary to experiment with the parameters of frequency resolution, N and δt, to obtain an unambiguous representation of the signal. Techniques for this are explored in Chapter 7.

Analysis of Signals in Frequency Space

Fourier analysis is a tool for extracting details about the frequencies that are present in a signal. Frequency analysis is routinely used in vibration analysis and isolation, determining the condition of bearings, and a wide variety of acoustic applications. An example from acoustics follows.

Example 2.8

Examine the frequency spectra of representative brass and woodwind instruments to illustrate why the characteristic sound of these instruments is so easily distinguished.

KNOWN Figures 2.20 and 2.21 provide a representative frequency spectrum for a clarinet, a woodwind instrument having a reed, and brass instruments.

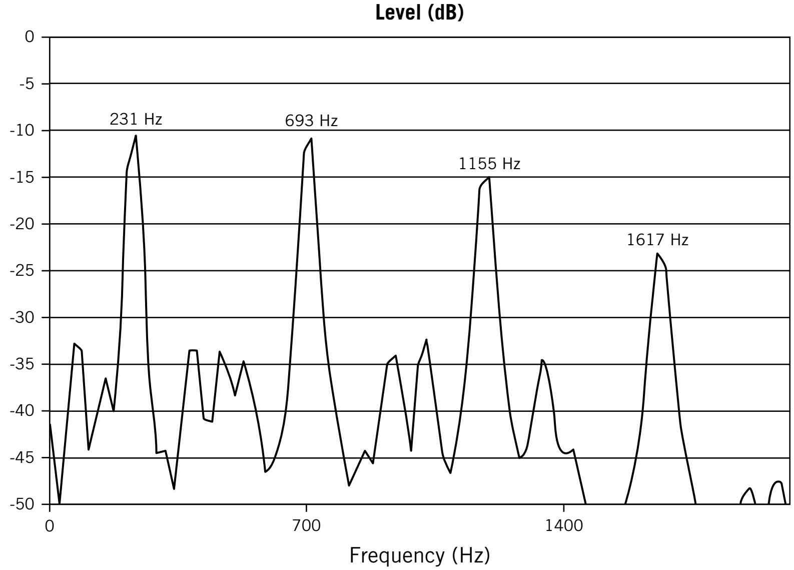


Figure 2.20 Frequency spectrum for a clarinet playing the note one whole tone below middle C. (Adapted from Campbell, D.M., Nonlinear Dynamics of Musical Reed and Brass Wind Instruments, *Contemporary Physics*, 40(6) November 1999, pages 415–431.)

DISCUSSION Figure 2.20 displays the frequency spectrum for a clarinet playing a B-flat. The base tone, f_o is at 231 Hz, with the harmonics having frequencies of $3f_o$, $5f_o$, etc. Contrast this with the frequency spectrum, shown in Figure 2.21, of the first few seconds of Aaron Copland's "Fanfare for the Common Man," which is played by an ensemble of brass instruments, like French horns, and has a fundamental frequency of approximately 700 Hz. This spectrum contains harmonics that are $2f_o$, $3f_o$, $4f_o$, etc.

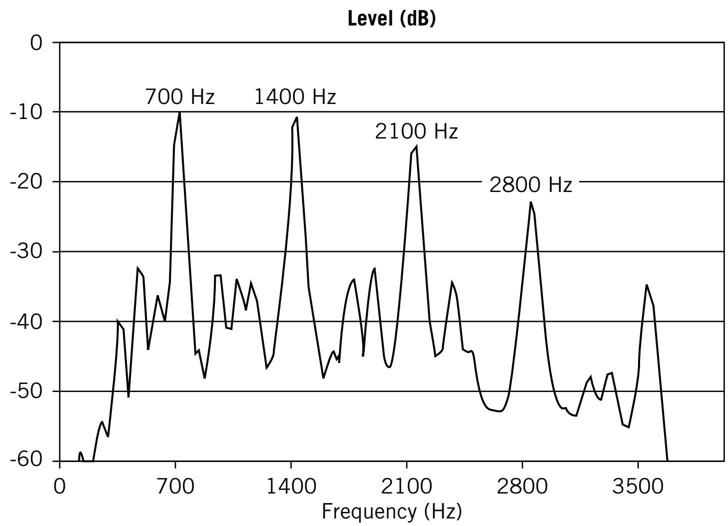


Figure 2.21 Frequency spectrum for the first three seconds of Aaron Copland's "Fanfare for the Common Man" that is played by the brass section of the orchestra.

COMMENT Fourier analysis and the examination of a signal's frequency content have myriads of applications. In this example, we demonstrated through frequency analysis a clear reason why clarinets and trumpets sound quite different. The presence of only odd harmonics, or both even and odd harmonics, is a very recognizable difference, easily perceived by the human ear.

2.6 SUMMARY

This chapter has provided a fundamental basis for the description of signals. The capabilities of a measurement system can be properly specified when the nature of the input signal is known. The descriptions of general classes of input signals will be seen in Chapter 3 to allow universal descriptions of measurement system dynamic behavior.

Any signal can be represented by a static magnitude and a series of varying frequencies and amplitudes. As such, measurement system selection and design must consider the frequency content of the input signals the system is intended to measure. Fourier analysis was introduced to allow a precise definition of the frequencies and the phase relationships among various frequency components within a particular signal. In Chapter 7, it will be shown as a tool for the accurate interpretation of discrete signals.

Signal characteristics form an important basis for the selection of measurement systems and the interpretation of measurement system output. In Chapter 3 these ideas are combined with the concept of a generalized set of measurement system behaviors. The combination of generalized measurement system behavior and generalized descriptions of input waveforms provides for an understanding of a wide range of instruments and measurement systems.

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3. Champeney, D. C., *Fourier Transforms and Their Physical Applications*, Academic, London, 1973.
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5. Cooley, J. W., and J. W. Tukey, An Algorithm for the Machine Calculation of Complex Fourier Series, *Mathematics of Computation* 19: 207, April 1965 (see also Special Issue on the fast Fourier transform, *IEEE Transactions on Audio and Electroacoustics* AU-2, June 1967).
6. Bendat, J. S.,and A. G. Piersol, *Random Data: Analysis and Measurement Procedures*, 3rd ed., Wiley, New York, 2000 (see also Bendat, J. S., and A. G. Piersol, *Engineering Applications of Correlation and Spectral Analysis*, 2nd ed., Wiley, New York, 1993).
7. Cochran, W. T., et al. What is the fast Fourier transform?, *Proceedings of the IEEE* 55(10): 1664, 1967.

SUGGESTED READING

Halliday, D.,and R. Resnick, *Fundamentals of Physics*, 6th ed., Wiley, New York, 2000.
Kreyszig, E., *Advanced Engineering Mathematics*, 9th ed., Wiley, New York, 2005.

NOMENCLATURE

f frequency, in Hz (t^{-1})
k spring constant (mt^{-2})
m mass (m)
t time (t)
y dependent variable
y_m discrete data points
A amplitude
B amplitude
C amplitude
F force (mlt^{-2})
N total number of discrete data points; integer
T period (t)
U unit step function
Y Fourier transform of y
α angle (rad)
β angle (rad)
δf frequency resolution (t^{-1})
δt sample time increment (t)
ϕ phase angle (rad)
ω circular frequency in rad/s (t^{-1})

PROBLEMS

2.1 Define the term "signal" as it relates to measurement systems. Provide two examples of static and dynamic input signals to particular measurement systems.

2.2 List the important characteristics of input and output signals and define each.

2.3 Determine the average and rms values for the function

$$y(t) = 25 + 10 \sin 6\pi t$$

over the time periods (a) 0 to 0.1 s, (b) 0.4 to 0.5 s, (c) 0 to 1/3 s, and (d) 0 to 20 s. Comment on the nature and meaning of the results in terms of analysis of dynamic signals.

2.4 The following values are obtained by sampling two time-varying signals once every 0.4 s:

t	$y_1(t)$	$y_2(t)$	t	$y_1(t)$	$y_2(t)$
0	0	0			
0.4	11.76	15.29	2.4	−11.76	−15.29
0.8	19.02	24.73	2.8	−19.02	−24.73
1.2	19.02	24.73	3.2	−19.02	−24.73
1.6	11.76	15.29	3.6	−11.76	−15.29
2.0	0	0	4.0	0	0

Determine the mean and the rms values for this discrete data. Discuss the significance of the rms value in distinguishing these signals.

2.5 A *moving average* is an averaging technique that can be applied to an analog, discrete time, or digital signal. A moving average is based on the concept of windowing, as illustrated in Figure 2.22. That portion of the signal that lies inside the window is averaged and the average values plotted as a function of time as the window moves across the signal. A 10-point moving average of the signal in Figure 2.22 is plotted in Figure 2.23.

a. Discuss the effects of employing a moving average on the signal depicted in Figure 2.22.

b. Develop a computer-based algorithm for computing a moving average, and determine the effect of the width of the averaging window on the signal described by

$$y(t) = \sin 5t + \cos 11t$$

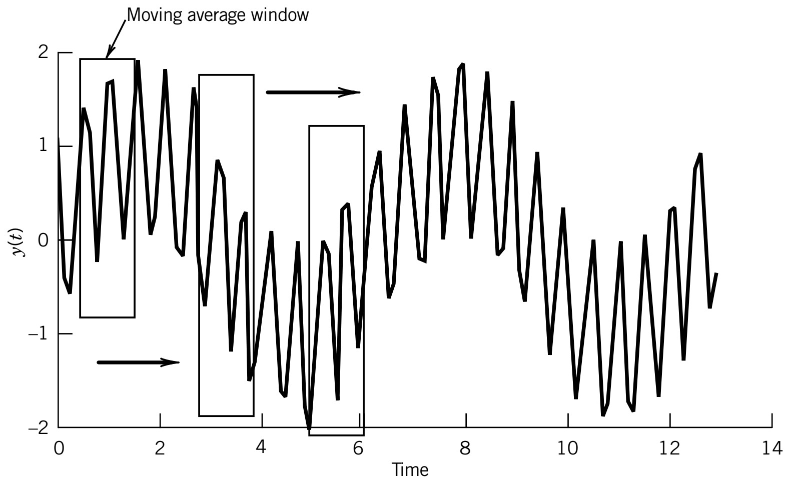

Figure 2.22 Moving average and windowing.

This signal should be represented as a discrete time signal by computing the value of the function at equally spaced time intervals. An appropriate time interval for this signal would be 0.05 s. Examine the signal with averaging windows of 4 and 30 points.

2.6 The data file in the companion software *noisy.txt* provides discrete time-varying signal that contains random noise. Apply a 2-, 3-, and 4-point moving average (see Problem 2.5) to these data, and plot the results. How does the moving average affect the noise in the data? Why?

2.7 Determine the value of the spring constant that would result in a spring-mass system that would execute one complete cycle of oscillation every 2.7 s, for a mass of 0.5 kg. What natural frequency does this system exhibit in radians/second?

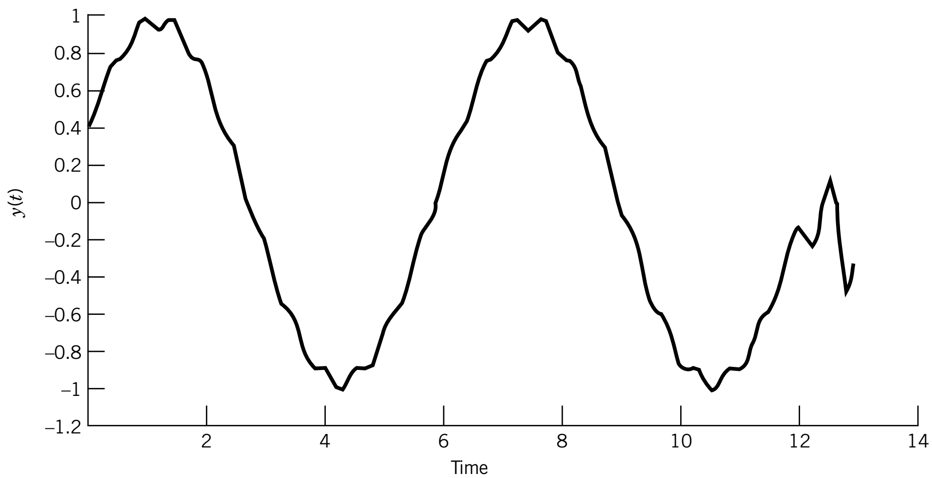

Figure 2.23 Effect of moving average on signal illustrated in Figure 2.22.

2.8 A spring with $k = 5000$ N/cm supports a mass of 1 kg. Determine the natural frequency of this system in radians/second and hertz.

2.9 For the following sine and cosine functions determine the period, the frequency in hertz, and the circular frequency in radians/second. (Note: t represents time in seconds).

a. $\sin 10\pi t/5$

b. $8 \cos 8t$

c. $\sin 5n\pi t$ for $n = 1$ to ∞

2.10 Express the following function in terms of (a) a cosine term only and (b) a sine term only:

$$y(t) = 5 \sin 4t + 3 \cos 4t$$

2.11 Express the function

$$y(t) = 4 \sin 2\pi t + 15 \cos 2\pi t$$

in terms of (a) a cosine term only and (b) a sine term only.

2.12 Express the Fourier series given by

$$y(t) = \sum_{n=1}^{\infty} \frac{2\pi n}{6} \sin n\pi t + \frac{4\pi n}{6} \cos n\pi t$$

using only cosine terms.

2.13 The nth partial sum of a Fourier series is defined as

$$A_0 + A_1 \cos \omega_1 t + B_1 \sin \omega_1 t + \cdots + \mathrm{A_n} \cos \omega_\mathrm{n} t + \mathrm{B_n} \sin \omega_\mathrm{n} t$$

For the third partial sum of the Fourier series given by

$$y(t) = \sum_{n=1}^{\infty} \frac{3n}{2} \sin nt + \frac{5n}{3} \cos nt$$

a. What is the fundamental frequency and the associated period?

b. Express this partial sum as cosine terms only.

2.14 For the Fourier series given by

$$y(t) = 4 + \sum_{n=1}^{\infty} \frac{2\pi n}{10} \cos \frac{n\pi}{4} t + \frac{120 n\pi}{30} \sin \frac{n\pi}{4} t$$

where t is time in seconds:

a. What is the fundamental frequency in hertz and radians/second?

b. What is the period T associated with the fundamental frequency?

c. Express this Fourier series as an infinite series containing sine terms only.

2.15 Find the Fourier series of the function shown in Figure 2.24, assuming the function has a period of 2π. Plot an accurate graph of the first three partial sums of the resulting Fourier series.

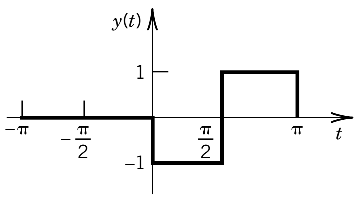

Figure 2.24 Function to be expanded in a Fourier series in Problem 2.15.

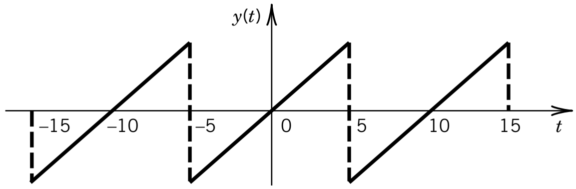

Figure 2.25 Sketch for Problem 2.16.

2.16 Determine the Fourier series for the function

$$y(t) = t \text{ for } -5 < t < 5$$

by expanding the function as an odd periodic function with a period of 10 units, as shown in Figure 2.25. Plot the first, second, and third partial sums of this Fourier series.

2.17 **a.** Show that $y(t) = t^2(-\pi < t < \pi)$, $y(t + 2\pi) = y(t)$ has the Fourier series

$$y(t) = \frac{\pi^2}{3} - 4\left(\cos t - \frac{1}{4}\cos 2t + \frac{1}{9}\cos 3t - + \cdots\right)$$

b. By setting $t = \pi$ in this series, show that a series approximation for π, first discovered by Euler, results as

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}$$

2.18 Determine the Fourier series that represents the function $y(t)$ where

$$y(t) = t \text{ for } 0 < t < 1$$

and

$$y(t) = 2 - t \text{ for } 0 < t < 1$$

Clearly explain your choice for extending the function to make it periodic.

2.19 Classify the following signals as static or dynamic, and identify any that may be classified as periodic:

a. $\sin 10t$ V

b. $5 + 2\cos 2t$ m

c. $5t$ s

d. 2 V

2.20 A particle executes linear harmonic motion around the point $x = 0$. At time zero the particle is at the point $x = 0$ and has a velocity of 5 cm/s. The frequency of the motion is 1 Hz. Determine: (a) the period, (b) the amplitude of the motion, (c) the displacement as a function of time, and (d) the maximum speed.

2.21 Define the following characteristics of signals: (a) frequency content, (b) amplitude, (c) magnitude, and (d) period.

2.22 Construct an amplitude spectrum plot for the Fourier series in Problem 2.16 for $y(t) = t$. Discuss the significance of this spectrum for measurement or interpretation of this signal. Hint: The plot can be done by inspection or by using software such as *DataSpect*.

2.23 Construct an amplitude spectrum plot for the Fourier series in Problem 2.17 for $y(t) = t^2$. Discuss the significance of this spectrum for selecting a measurement system. Hint: The plot can be done by inspection or by using software such as *DataSpect* or *WaveformGeneration.vi.*

2.24 Sketch representative waveforms of the following signals, and represent them as mathematical functions (if possible):

a. The output signal from the thermostat on a refrigerator.

b. The electrical signal to a spark plug in a car engine.

c. The input to a cruise control from an automobile.

d. A pure musical tone (e.g., 440 Hz is the note A).

e. The note produced by a guitar string.

f. AM and FM radio signals.

2.25 Represent the function

$$e(t) = 5 \sin 31.4t + 2 \sin 44t$$

as a discrete set of $N = 128$ numbers separated by a time increment of $(1/N)$. Use an appropriate algorithm to construct an amplitude spectrum from this data set. (Hint: A spreadsheet program or the program file *DataSpect* will handle this task.)

2.26 Repeat Problem 2.25 using a data set of 256 numbers at $\delta t = (1/N)$ and $\delta t = (1/2N)$ seconds. Compare and discuss the results.

2.27 A particular strain sensor is mounted to an aircraft wing that is subjected to periodic wind gusts. The strain measurement system indicates a periodic strain that ranges from 3250×10^{-6} in./in. to 4150×10^{-6} in./in. at a frequency of 1 Hz. Determine:

a. The average value of this signal.

b. The amplitude and the frequency of this output signal when expressed as a simple periodic function.

c. A one-term Fourier series that represents this signal.

d. Construct an amplitude spectrum plot for the output signal.

2.28 For a dynamic calibration involving a force measurement system, a known force is applied to a sensor. The force varies between 100 and 170 N at a frequency of 10 rad/s. State the average (static) value of the input signal, its amplitude, and its frequency. Assuming that the signal may be represented by a simple periodic waveform, express the signal as a one-term Fourier series and create an amplitude spectrum from an equivalent discrete time series.

2.29 A displacement sensor is placed on a dynamic calibration rig known as a *shaker*. This device produces a known periodic displacement that serves as the input to the sensor. If the known displacement is set to vary between 2 and 5 mm at a frequency of 100 Hz, express the input signal as a one-term Fourier series. Plot the signal in the time domain, and construct an amplitude spectrum plot.

2.30 Consider the upward flow of water and air in a tube having a circular cross section, as shown in Figure 2.26. If the water and air flow rates are within a certain range, there are slugs of liquid and large gas bubbles flowing upward together. This type of flow is called "slug flow." The data file *gas_liquid_data.txt* with the companion software contains measurements of pressure made at the wall of a tube in which air and water were flowing. The data were acquired at a sample frequency of 300 Hz. The average flow velocity of the air and water is 1 m/s.

a. Construct an amplitude spectrum from the data, and determine the dominant frequency.

b. Using the frequency information from part **a**, determine the length L shown in the drawing in Figure 2.26. Assume that the dominant frequency is associated with the passage of the bubbles and slugs across the pressure sensor.

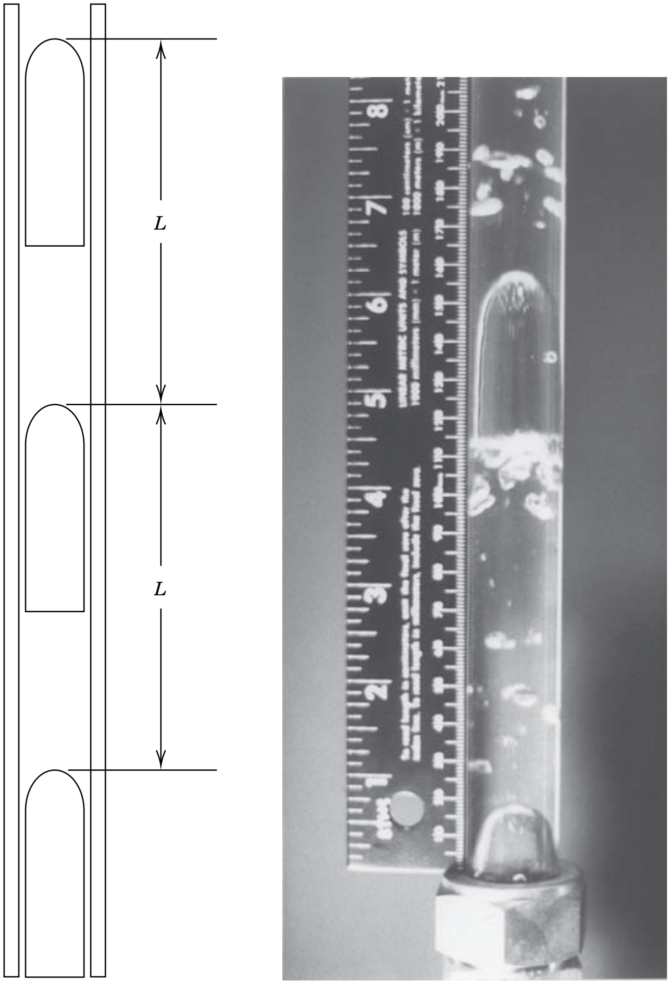

Figure 2.26 Upward gas–liquid flow: slug flow regime.

2.31 The file *sunspot.txt* contains data from monthly observations of sunspot activity from January 1746 to September 2005. The numbers are a relative measure of sunspot activity.

a. Using the companion software program Dataspect, plot the data and create an amplitude spectrum.

b. From the spectrum, identify any cycles present in the sunspot activity and determine the period(s).

c. The Dalton minimum describes a period of low sunspot activity lasting from about 1790 to 1830; this is apparent in your plot of the data. The Dalton minimum occurred near the end of the Little Ice Age and coincided with a period of lower-than-average global temperatures. Research the ‘‘year without a summer’’ and its relationship to the Dalton minimum.

2.32 Classify the following signals as completely as possible:

a. Clock face having hands.

b. Morse code.

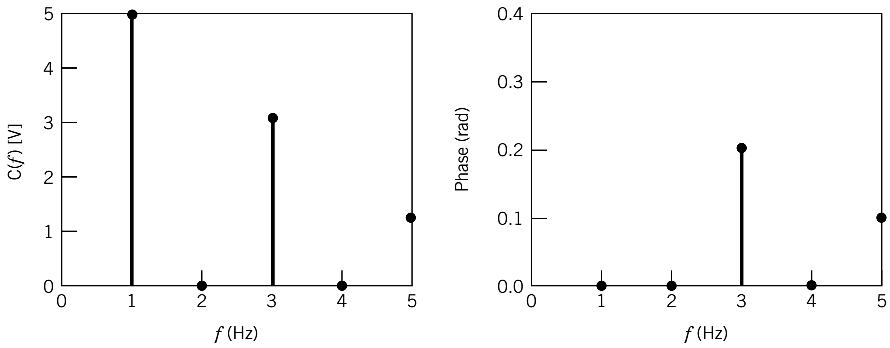

Figure 2.27 Spectrum for Problem 2.33.

c. Musical score, as input to a musician.

d. Flashing neon sign.

e. Telephone conversation.

f. Fax transmission.

2.33 Describe the signal defined by the amplitude spectrum and its phase shift of Figure 2.27 in terms of its Fourier series. What is the frequency resolution of these plots? What was the sample time increment used?

2.34 For the even-functioned triangle wave signal defined by

$$y(t) = (4C/T)t + C \quad -T/2 \leq t \leq 0$$
$$y(t) = (-4C/T)t + C \quad 0 \leq t \leq T/2$$

where C is an amplitude and T is the period:

a. Show that this signal can be represented by the Fourier series

$$y(t) = \sum_{n=1}^{\infty} \frac{4C(1 - \cos n\pi)}{(\pi n)^2} \cos \frac{2\pi n t}{T}$$

b. Expand the first three nonzero terms of the series. Identify the terms containing the fundamental frequency and its harmonics.

c. Plot each of the first three terms on a separate graph. Then plot the sum of the three. For this, set $C = 1$ V and $T = 1$ s. Note: The program *FourCoef* or *Waveform Generation* is useful in this problem.

d. Sketch the amplitude spectrum for the first three terms of this series, first separately for each term and then combined.

2.35 Figure 2.15 illustrates how the inclusion of higher frequency terms in a Fourier series refine the accuracy of the series representation of the original function. However, measured signals often display an amplitude spectrum for which amplitudes decrease with frequency. Using Figure 2.15 as a resource, discuss the effect of high-frequency, low-amplitude noise on a measured signal. What signal characteristics would be unaffected? Consider both periodic and random noise.

2.36 The program *Sound.vi* samples the ambient room sounds using your laptop computer microphone and sound board and returns the amplitude spectrum of the sounds. Experiment with different sounds (e.g., tapping, whistling, talking, humming) and describe your findings in a brief report.